**SQL QUESTIONS**

**1 Avg engagement by credibility group**

```
SELECT

  CASE

    WHEN LOWER(TRIM(Rating)) IN ('true','no factual content')

    THEN 'Human_Content'

    ELSE 'AI_Like_Content'

  END AS content_group,



  AVG(reaction_count + comment_count + share_count) AS avg_engagement



FROM facebook

GROUP BY content_group;
```

**RESULT:**

| | content_group | avg_engagement |
|---|---|---|
| 1 | AI_Like_Content | 6076.82397572079 |
| 2 | Human_Content | 42771.2765957447 |

**2. Total posts in each group**

```
SELECT

  CASE

    WHEN LOWER(TRIM(Rating)) IN ('true','no factual content')

    THEN 'Human_Content'
```

```
    ELSE 'AI_Like_Content'
  END AS content_group,
  COUNT(*) AS total_posts
FROM facebook
GROUP BY content_group;
```

**RESULT:**

| | content_group | total_posts |
|---|---|---|
| 1 | AI_Like_Content | 2018 |
| 2 | Human_Content | 264 |

**3. Overall average engagement**

```
SELECT
  AVG(reaction_count + comment_count + share_count) AS overall_avg_engagement
FROM facebook;
```

**RESULT**

| | overall_avg_engagement |
|---|---|
| 1 | 9975.19484629295 |

Engagement by Page (publisher)

```
SELECT
```

```
  Page,

  AVG(reaction_count + comment_count + share_count) AS avg_engagement

FROM facebook

GROUP BY Page

ORDER BY avg_engagement DESC;
```

**ANSWER:**

| | Page | avg_engagement |
|---|---|---|
| 1 | Occupy Democrats | 67340.6815920398 |
| 2 | The Other 98% | 42144.2547169811 |
| 3 | Freedom Daily | 6675.72072072072 |
| 4 | Addicting Info | 4818.92537313433 |
| 5 | Right Wing News | 4221.84269662921 |
| 6 | Politico | 1256.02621722846 |
| 7 | Eagle Rising | 1229.31046931408 |

5. Engagement by Category

```
SELECT

  Category,

  AVG(reaction_count + comment_count + share_count) AS avg_engagement

FROM facebook

GROUP BY Category

ORDER BY avg_engagement DESC;
```

**RESULT:**

| | Category | avg_engagement |
|---|---|---|
| 1 | left | 42286.8571428571 |
| 2 | right | 3372.1465648855 |
| 3 | mainstream | 1082.32168458781 |

6. Engagement by Post Type

```
SELECT
  "Post Type",
  AVG(reaction_count + comment_count + share_count) AS avg_engagement
FROM facebook
GROUP BY "Post Type"
ORDER BY avg_engagement DESC;
```

**RESULT:**

| | Post Type | avg_engagement |
|---|---|---|
| 1 | photo | 42613.6275510204 |
| 2 | video | 27594.1926229508 |
| 3 | link | 3947.69400452489 |
| 4 | text | 88.5 |

**7.Top 10 most engaging posts**

```
SELECT
  post_id,
  Page,
  "Post URL",
  (reaction_count + comment_count + share_count) AS total_engagement
FROM facebook
ORDER BY total_engagement DESC
LIMIT 10;
```

**RESULT:**

| | post_id | Page | Post URL | total_engagement |
|---|---|---|---|---|
| 1 | 1247448632014855 | Occupy Democrats | https://www.facebook.com/OccupyDemocrats/posts/... | 1704500 |
| 2 | 1246675775425474 | Occupy Democrats | https://www.facebook.com/OccupyDemocrats/posts/... | 673046 |
| 3 | 1253304291429289 | Occupy Democrats | https://www.facebook.com/OccupyDemocrats/posts/... | 546154 |
| 4 | 1253151444777907 | Occupy Democrats | https://www.facebook.com/OccupyDemocrats/posts/... | 491157 |
| 5 | 1249740458452339 | Occupy Democrats | https://www.facebook.com/OccupyDemocrats/posts/... | 487206 |
| 6 | 1471293542881619 | The Other 98% | https://www.facebook.com/TheOther98/posts/... | 434467 |
| 7 | 1252823384810713 | Occupy Democrats | https://www.facebook.com/OccupyDemocrats/posts/... | 421505 |

**8.Avg shares by credibility group**

```
SELECT
  CASE
    WHEN LOWER(TRIM(Rating)) IN ('true','no factual content')
    THEN 'Human_Content'
    ELSE 'AI_Like_Content'
  END AS content_group,
  AVG(share_count) AS avg_shares
FROM facebook
GROUP BY content_group;
```

**ANSWER:**

| | content_group | avg_shares |
|---|---|---|
| 1 | AI_Like_Content | 2190.0799190693 |
| 2 | Human_Content | 19648.2723404255 |

**9.Avg comments by credibility group**

```
SELECT
  CASE
    WHEN LOWER(TRIM(Rating)) IN ('true','no factual content')
    THEN 'Human_Content'
    ELSE 'AI_Like_Content'
  END AS content_group,
  AVG(comment_count) AS avg_comments
FROM facebook
GROUP BY content_group;
```

**RESULT:**

| | content_group | avg_comments |
|---|---|---|
| 1 | AI_Like_Content | 391.283234126984 |
| 2 | Human_Content | 1469.26515151515 |

**10.Posts with zero engagement**

```
SELECT COUNT(*) AS zero_engagement_posts

FROM facebook

WHERE (reaction_count + comment_count + share_count) = 0;
```

**RESULT:**

| | zero_engagement_posts |
|---|---|
| 1 | 0 |

**11.Highest discussion posts (top comments)**

```
SELECT

  post_id,

  Page,

  comment_count

FROM facebook

ORDER BY comment_count DESC

LIMIT 10;
```

**RESULT:**

| | post_id | Page | comment_count |
|---|---|---|---|
| 1 | 1247448632014855 | Occupy Democrats | 159047 |
| 2 | 1246675775425474 | Occupy Democrats | 32419 |
| 3 | 1252823384810713 | Occupy Democrats | 20576 |
| 4 | 1249740458452339 | Occupy Democrats | 13657 |
| 5 | 1472059186138388 | The Other 98% | 13530 |
| 6 | 1245835812176137 | Occupy Democrats | 13510 |
| 7 | 1245838695509182 | Occupy Democrats | 13510 |

**12.Engagement by weekday**

```
SELECT

  STRFTIME('%w', "Date Published") AS weekday,

  AVG(reaction_count + comment_count + share_count) AS avg_engagement

FROM facebook

GROUP BY weekday;
```

| | weekday | avg_engagement |
|---|---|---|
| 1 | 1 | 7681.08516886931 |
| 2 | 2 | 13259.5615615616 |
| 3 | 3 | 12449.4983050847 |
| 4 | 4 | 6174.57543859649 |
| 5 | 5 | 9021.36140350877 |

**13.% engagement driven by AI-like content**

```
SELECT

  SUM(CASE WHEN LOWER(TRIM(Rating)) NOT IN ('true','no factual content')

          THEN (reaction_count + comment_count + share_count)

          ELSE 0 END) * 1.0

  / SUM(reaction_count + comment_count + share_count) AS ai_engagement_share

FROM facebook;
```

**RESULT:**

| | ai_engagement_share |
|---|---|
| 1 | 0.544473585948799 |

**–14.Engagement buckets (Low/Medium/High)**

```
SELECT

  CASE

    WHEN (reaction_count + comment_count + share_count) < 50 THEN 'Low'

    WHEN (reaction_count + comment_count + share_count) < 200 THEN 'Medium'

    ELSE 'High'

  END AS engagement_bucket,

  COUNT(*) AS posts

FROM facebook

GROUP BY engagement_bucket;
```

**RESULT:**

| | engagement_bucket | posts |
|---|---|---|
| 1 | High | 1812 |
| 2 | Low | 86 |
| 3 | Medium | 384 |

**–15 High-risk viral misinformation posts**

```
SELECT
  post_id,
  Page,
  Rating,
  (share_count * 1.0 / (reaction_count + 1)) AS virality_score
FROM facebook
WHERE LOWER(TRIM(Rating)) NOT IN ('true','no factual content')
  AND (share_count * 1.0 / (reaction_count + 1)) > 1
ORDER BY virality_score DESC;
```

| | post_id | Page | Rating | virality_score |
|---|---|---|---|---|
| 1 | 552582351611944 | Eagle Rising | mixture of true and false | 3.82707774798928 |
| 2 | 1282890125086151 | CNN Politics | mostly true | 3.31594202898551 |
| 3 | 552599694943543 | Eagle Rising | mixture of true and false | 3.10542115486491 |
| 4 | 1472059186138388 | The Other 98% | mostly true | 2.87881225374579 |
| 5 | 1249978638397584 | Freedom Daily | mostly true | 2.74669573511378 |
| 6 | 551828675020645 | Eagle Rising | mostly false | 2.6109022556391 |
| 7 | 552487701631409 | Eagle Rising | mostly false | 2.6109022556391 |

**16.Risky content count by Page**

```
SELECT
  Page,
  COUNT(*) AS risky_posts
FROM facebook
WHERE LOWER(TRIM(Rating)) NOT IN ('true','no factual content')
GROUP BY Page
ORDER BY risky_posts DESC;
```

```
ORDER BY risky_posts DESC;
```

| | Page | risky_posts |
|---|---|---|
| 1 | Politico | 530 |
| 2 | CNN Politics | 389 |
| 3 | Right Wing News | 257 |
| 4 | Eagle Rising | 205 |
| 5 | ABC News Politics | 174 |
| 6 | Occupy Democrats | 144 |
| 7 | Addicting Info | 129 |

**17 Engagement per post (efficiency).**

| | Page | engagement_per_post |
|---|---|---|
| 1 | ABC News Politics | 278.465 |
| 2 | Addicting Info | 4612.4 |
| 3 | CNN Politics | 1177.16381418093 |
| 4 | Eagle Rising | 1190.62587412587 |
| 5 | Freedom Daily | 6616.11607142857 |
| 6 | Occupy Democrats | 64763.04784689 |
| 7 | Politico | 1251.33955223881 |

**18.Monthly engagement trend**

```
SELECT
  STRFTIME('%Y-%m', "Date Published") AS month,
  AVG(reaction_count + comment_count + share_count) AS avg_engagement
FROM facebook
GROUP BY month
ORDER BY month;
```

| | month | avg_engagement |
|---|---|---|
| 1 | 2016-09 | 9975.19484629295 |

**19.Monthly volume of risky content**

```
SELECT

  STRFTIME('%Y-%m', "Date Published") AS month,

  COUNT(*) AS risky_posts

FROM facebook

WHERE LOWER(TRIM(Rating)) NOT IN ('true','no factual content')

GROUP BY month;
```

| | month | risky_posts |
|---|---|---|
| 1 | 2016-09 | 2018 |

**20.Debate vs engagement relationship**

```
SELECT

  Debate,

  AVG(reaction_count + comment_count + share_count) AS avg_engagement

FROM facebook

GROUP BY Debate;
```

| | Debate | avg_engagement |
|---|---|---|
| 1 | NULL | 9053.63333333333 |
| 2 | yes | 16034.7773972603 |

**21.Posts with high engagement but low virality**

```
SELECT

  post_id,

  Page

FROM facebook

WHERE (reaction_count + comment_count + share_count) > 200

  AND (share_count * 1.0 / (reaction_count + 1)) < 0.2;
```

| | post_id | Page |
|---|---|---|
| 1 | 1035352946562931 | ABC News Politics |
| 2 | 1035411183223774 | ABC News Politics |
| 3 | 1035491416549084 | ABC News Politics |
| 4 | 1035592746538951 | ABC News Politics |
| 5 | 1035595553205337 | ABC News Politics |
| 6 | 1035619779869581 | ABC News Politics |
| 7 | 1035706656527560 | ABC News Politics |

22 . Rank posts within each Rating group

```
SELECT
  post_id,
  Rating,
  (reaction_count + comment_count + share_count) AS total_engagement,
  RANK() OVER (
    PARTITION BY Rating
    ORDER BY (reaction_count + comment_count + share_count) DESC
  ) AS rank_in_rating
FROM facebook;
```

| | post_id | Rating | total_engagement | rank_in_rating |
|---|---|---|---|---|
| 1 | 1246675775425474 | mixture of true and false | 673046 | 1 |
| 2 | 1247777691981949 | mixture of true and false | 180355 | 2 |
| 3 | 1253904648035920 | mixture of true and false | 171990 | 3 |
| 4 | 1245894178836967 | mixture of true and false | 80209 | 4 |
| 5 | 1252629544830097 | mixture of true and false | 74267 | 5 |
| 6 | 1246126732147045 | mixture of true and false | 66604 | 6 |
| 7 | 10153842821881680 | mixture of true and false | 57205 | 7 |
| 8 | 1636717173155073 | mixture of true and false | 56403 | 8 |

23. Top 10% engagement posts

```
SELECT *
FROM (
  SELECT
    post_id,
    (reaction_count + comment_count + share_count) AS total_engagement,
    PERCENT_RANK() OVER (
      ORDER BY (reaction_count + comment_count + share_count)
    ) AS pr
  FROM facebook
)
WHERE pr >= 0.9;
```

RESULTS:

| | post_id | total_engagement | pr |
|---|---|---|---|
| 1 | 1449004081806726 | 18758 | 0.900043840420868 |
| 2 | 1252884128137972 | 18774 | 0.900482244629548 |
| 3 | 1466093066735000 | 19007 | 0.900920648838229 |
| 4 | 1452117881495346 | 19015 | 0.901359053046909 |
| 5 | 1253200038106381 | 19541 | 0.90179745725559 |
| 6 | 1245754342184284 | 19621 | 0.90223586146427 |
| 7 | 1455575034482964 | 19967 | 0.902674265672951 |
| 8 | 1249124161816365 | 20003 | 0.903112669881631 |

24.Engagement share contribution per post

```
SELECT
  post_id,
  (reaction_count + comment_count + share_count) AS total_engagement,
  (reaction_count + comment_count + share_count) * 1.0 /
  SUM(reaction_count + comment_count + share_count) OVER () AS engagement_share
```

FROM facebook;

| | post_id | total_engagement | engagement_share |
|---|---|---|---|
| 1 | 1035057923259100 | NULL | NULL |
| 2 | 1035269309904628 | 68 | 3.08178546503984e-06 |
| 3 | 1035305953234297 | 124 | 5.61972643624912e-06 |
| 4 | 1035322636565962 | 291 | 1.31882289753911e-05 |
| 5 | 1035352946562931 | 6571 | 0.000297800180746718 |
| 6 | 1035366579894901 | 72 | 3.26306696298336e-06 |
| 7 | 1035411183223774 | 560 | 2.53794097120928e-05 |
| 8 | 1035430716555154 | 133 | 6.02760980662204e-06 |

–25. **Rolling 7-day engagement trend**

```
SELECT
  DATE("Date Published") AS day,
  AVG(reaction_count + comment_count + share_count) AS daily_avg,
  AVG(AVG(reaction_count + comment_count + share_count)) OVER (
    ORDER BY DATE("Date Published")
    ROWS BETWEEN 6 PRECEDING AND CURRENT ROW
  ) AS rolling_7_day_avg
FROM facebook
GROUP BY day;
```

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| | day | daily_avg | rolling_7_day_avg |
|---|---|---|---|
| 1 | 2016-09-19 | 5327.95622895623 | 5327.95622895623 |
| 2 | 2016-09-20 | 11148.4466019417 | 8238.20141544899 |
| 3 | 2016-09-21 | 12449.4983050847 | 9641.96704532757 |
| 4 | 2016-09-22 | 6174.57543859649 | 8775.1191436448 |
| 5 | 2016-09-23 | 9021.36140350877 | 8824.3675956176 |
| 6 | 2016-09-26 | 9501.08333333333 | 8937.15355190355 |
| 7 | 2016-09-27 | 15086.8291316527 | 9815.67863472485 |

26 . Engagement quartiles

```
SELECT
  post_id,
  NTILE(4) OVER (
    ORDER BY (reaction_count + comment_count + share_count) DESC
  ) AS engagement_quartile
FROM facebook;
```

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| | post_id | engagement_quartile |
|---|---|---|
| 1 | 1247448632014855 | 1 |
| 2 | 1246675775425474 | 1 |
| 3 | 1253304291429289 | 1 |
| 4 | 1253151444777907 | 1 |
| 5 | 1249740458452339 | 1 |
| 6 | 1471293542881619 | 1 |
| 7 | 1252823384810713 | 1 |
| 8 | 1463581596986147 | 1 |

26.**Top post per Page**

```
SELECT *
FROM (
  SELECT
    post_id,
    Page,
    (reaction_count + comment_count + share_count) AS total_engagement,
    RANK() OVER (
      PARTITION BY Page
      ORDER BY (reaction_count + comment_count + share_count) DESC
    ) AS page_rank
  FROM facebook
)
WHERE page_rank = 1;
```

| | post_id | Page | total_engagement | page_rank |
|---|---|---|---|---|
| 1 | 1035352946562931 | ABC News Politics | 6571 | 1 |
| 2 | 1451127358261065 | Addicting Info | 27580 | 1 |
| 3 | 1282890125086151 | CNN Politics | 30924 | 1 |
| 4 | 552582351611944 | Eagle Rising | 32105 | 1 |
| 5 | 1249978638397584 | Freedom Daily | 55850 | 1 |
| 6 | 1247448632014855 | Occupy Democrats | 1704500 | 1 |
| 7 | 10153842821881680 | Politico | 57205 | 1 |

28.High-risk posts inside top engagement tier

```
SELECT *
FROM (
  SELECT
    post_id,
    Rating,
    (reaction_count + comment_count + share_count) AS total_engagement,
    NTILE(3) OVER (
      ORDER BY (reaction_count + comment_count + share_count) DESC
    ) AS tier
  FROM facebook
)
WHERE tier = 1
  AND LOWER(TRIM(Rating)) NOT IN ('true','no factual content');
```

| | post_id | Rating | total_engagement | tier |
|---|---|---|---|---|
| 1 | 1246675775425474 | mixture of true and false | 673046 | 1 |
| 2 | 1253304291429289 | mostly true | 546154 | 1 |
| 3 | 1472059186138388 | mostly true | 298036 | 1 |
| 4 | 1472225646121742 | mostly true | 223544 | 1 |
| 5 | 1247777691981949 | mixture of true and false | 180355 | 1 |
| 6 | 1252892884803763 | mostly true | 174098 | 1 |
| 7 | 1253904648035920 | mixture of true and false | 171990 | 1 |
| 8 | 1252897674803284 | mostly true | 171855 | 1 |

-29Rank posts by engagement within each Page

```
SELECT
  post_id,
  Page,
  (reaction_count + comment_count + share_count) AS total_engagement,
```

```
  RANK() OVER (
    PARTITION BY Page
    ORDER BY (reaction_count + comment_count + share_count) DESC
  ) AS rank_on_page
FROM facebook;
```

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| | post_id | Page | total_engagement | rank_on_page |
|---|---|---|---|---|
| 1 | 1035352946562931 | ABC News Politics | 6571 | 1 |
| 2 | 1042051285893097 | ABC News Politics | 2754 | 2 |
| 3 | 1041100572654835 | ABC News Politics | 2251 | 3 |
| 4 | 1042083579223201 | ABC News Politics | 1685 | 4 |
| 5 | 1041240549307504 | ABC News Politics | 1406 | 5 |
| 6 | 1037097846388441 | ABC News Politics | 1336 | 6 |
| 7 | 1041257462639146 | ABC News Politics | 1296 | 7 |
| 8 | 1035656723199220 | ABC News Politics | 1206 | 8 |

30.High-risk posts inside top engagement tier

```
WITH ranked_posts AS (
  SELECT
    post_id,
    Rating,
    (reaction_count + comment_count + share_count) AS total_engagement,
    NTILE(3) OVER (
      ORDER BY (reaction_count + comment_count + share_count) DESC
    ) AS tier
  FROM facebook
)

SELECT *
FROM ranked_posts
WHERE tier = 1
  AND LOWER(TRIM(Rating)) NOT IN ('true','no factual content');
```

| | post_id | Rating | total_engagement | tier |
|---|---|---|---|---|
| 1 | 1246675775425474 | mixture of true and false | 673046 | 1 |
| 2 | 1253304291429289 | mostly true | 546154 | 1 |
| 3 | 1472059186138388 | mostly true | 298036 | 1 |
| 4 | 1472225646121742 | mostly true | 223544 | 1 |
| 5 | 1247777691981949 | mixture of true and false | 180355 | 1 |
| 6 | 1252892884803763 | mostly true | 174098 | 1 |
| 7 | 1253904648035920 | mixture of true and false | 171990 | 1 |

31. Publisher performance above platform average

```
WITH platform_avg AS (
  SELECT
    AVG(reaction_count + comment_count + share_count) AS avg_engagement
  FROM facebook
)

SELECT
  Page,
  AVG(reaction_count + comment_count + share_count) AS page_avg
FROM facebook, platform_avg
GROUP BY Page
HAVING page_avg > avg_engagement;
```

| | Page | page_avg |
|---|---|---|
| 1 | Occupy Democrats | 67340.6815920398 |
| 2 | The Other 98% | 42144.2547169811 |

—32 Monthly misinformation engagement share

```
WITH monthly AS (
  SELECT
    STRFTIME('%Y-%m', "Date Published") AS month,
    SUM(reaction_count + comment_count + share_count) AS total_engagement,
    SUM(CASE
        WHEN LOWER(TRIM(Rating)) NOT IN ('true','no factual content')
        THEN (reaction_count + comment_count + share_count)
        ELSE 0
      END) AS misleading_engagement
```

```
  FROM facebook
  GROUP BY month
)

SELECT
  month,
  misleading_engagement * 1.0 / total_engagement AS misinformation_share
FROM monthly
ORDER BY month;
```

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| | month | misinformation_share |
|---|---|---|
| 1 | 2016-09 | 0.5444735859487 99 |